Education

Institute of Technology, Nirma University Ahmedabad, India
Bachelor in Computer Science and Technology *2021 – 2025*

- Cumulative GPA : **7.97/10.0**
- Relevant Coursework: Machine Learning, Deep Learning, Probability, and Applied Statistics, Advance Data Structure and Algorithms, Natural Language Processing, Operating System, Database Management System, Computer Network, Big Data analytics, Compiler Construction, Blockchain Technology

Technical Skills

- **Programming Languages**: Python, C/C++, JavaScript, GO
- **Data Science and Miscellaneous Technologies**: Data Science Pipeline(collection, cleansing, visualization, modeling, interpretation)
- **Big Data**: : Hadoop, MapReduce, Apache Hive, Basics of Spark, Linux
- **Collaboration Tools**: Git, GitHub
- **Automation**: Docker
- **API and Web Services**: REST, Postman, FastAPI
- **ETL and Data Engineering**: Snowflake, Airflow, dbt (Data Build Tool)
- **Cloud Services**: AWS
- **Database Technologies**: PostgreSQL, MongoDB

Publications

- **Title:** A Faster R-CNN Approach For Bone Fracture Detection in Telesurgery Systems in Healthcare 4.0
- **Conference:** GCAT, 2024
- **Abstract:** This paper introduces an ideal structure for the detection and treatment of bone fractures utilizing advanced deep learning and telesurgery techniques. The approach begins with the collection of medical imaging data from the X-Ray Layer, which is subsequently divided into training, validation, and testing datasets. The proposed model, ResNeXt101, is trained on training dataset and validated with the validation dataset to ensure robust performance.

Projects

Medical-RAG | *Natural Language Processing*

- Developed a Retrieval-Augmented Generation (RAG) Chatbot powered by fine-tuned Language Models (LLM) and Semantic Chunking.
- Mitigated LLM hallucinations with a localized setup on CPU, ensuring reliable, personalized medical information delivery.
- Utilized Qdrant Docker database for efficient storage and retrieval of semantic chunks from medical documents.

DefectNet : Ensemble Learning Based Software Defect detection for Python Development | *Python*

- Developed DefectNet as python-centric mechanism for addressing JIT and line-based chronological order bugs.
- Designed an ensemble-learning model containing different machine learning Algorithms such as Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, Xgboost Classifier, Multinomial Naive Bayes and KNN.

AWS Unstructured Streaming | *Data Engineering*

- This project demonstrates a scalable approach for real-time processing of unstructured data using Apache Spark and AWS services.
- The application streams unstructured data, processes it to extract relevant details, and saves it in a structured format on AWS S3.

Zillow ETL Pipeline | *Data Engineering*

- Developed an end-to-end ETL pipeline to extract real estate data from Zillow/RapidAPI, store it in an S3 landing bucket, and process it using AWS Lambda.
- Transformed data is stored in an S3 bucket and loaded into Amazon Redshift for efficient querying and analysis.
- Integrated AWS QuickSight for interactive data visualization and insights to support better decision-making.

Data Streaming with Kafka | *Data Engineering*

- This project monitors a YouTube playlist using Confluent Kafka, Google API, and ksqlDB, and triggers real-time notifications via a Telegram bot when changes are detected (such as added likes,comments).

Certifications

Unsupervised Learning, Recommenders, Reinforcement Learning | Coursera, Oct 2023

- **Key Skills**: Reinforcement Learning, Recommender Systems, Unsupervised Learning, Anomaly Detection

Advanced Learning Algorithms | Coursera, Sept 2023

- **Key Skills**: Artificial Neural Networks, TensorFlow, XGBoost, Model Development, Python, Data Preprocessing, Model Optimization, Evaluation